

How Much Damage Can GenAI Repair? Fact Recovery in Damaged Historical Newspaper Text

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Abstract

[TODO: Write abstract after results are final.]

1 Introduction

Digitised newspapers are now a core source for historical research. Digitisation lets scholars search and mine large collections by keyword rather than reading page by page (Nguyen et al., 2021). Yet what historians search and mine is not the page itself but the text that an OCR engine read from a scan of it. OCR errors measurably change the results of quantitative analyses (Hill and Hengchen, 2019) and harm downstream NLP tasks (Strien et al., 2020), with named entities among the most affected (Hamdi et al., 2023). For 19th-century British newspapers the mean character error rate (CER) is about 7%, but it ranges from 0.3% to 44.5% between excerpts of a single title, and this spread tracks the condition of the source page, not just the OCR engine: faded or physically damaged originals give the engine the least to work with (Booth et al., 2024). For the British Library’s own 19th-century newspaper archive, character accuracy reaches 83.6% but word accuracy falls to 78%, and accuracy for capitalised words and number groups drops further still, to 63.4% and 64.1% (Tanner et al., 2009). The words hit hardest are thus often the facts a historian searches for: names, places, dates and sums of money.

Generative AI makes it tempting to repair such text automatically, but the evidence is mixed. Llama 2, instruction-tuned on BLN600, reduced the CER of its OCR by 54.5%, against 23.3% for a fine-tuned BART (Thomas et al., 2024), while a study of 14 LLMs on several historical benchmarks found them “anything but efficient” at post-correction (Boros et al., 2024). There is also a deeper risk. A generative model always returns fluent text, even where the evidence for a word is gone, and such models are known to produce content that

their input does not support (Ji et al., 2023). For an archive, a plausible but wrong name is worse than visible noise, because the reader can no longer see that anything was damaged. Similarity-based metrics may hide this error too: a sentence with the wrong name can still score as close in meaning to the original.

Existing post-OCR correction studies score whole documents at whatever error level the OCR happened to produce. They show that a model helps on average, but not how far repair can be trusted as damage grows, nor whether the facts inside the damaged region survive. We therefore ask:

RQ. When the damaged words of a sentence are known and their garbled letters are visible, how much of the original wording, and in particular its names, numbers and dates, can a fine-tuned masked language model and a few-shot instruction-following LLM restore, and how does this change as the damaged share of the sentence grows?

We split it into three sub-questions. **SQ1** (severity): does repair quality fall smoothly as damage grows, or break down at some level? **SQ2** (method): given identical input, does a fine-tuned masked language model or a few-shot LLM repair better? **SQ3** (metric validity): does semantic similarity overstate repair quality compared with exact fact recovery?

Answering these questions needs damage whose amount we control and whose original text we know, which real damaged pages cannot give. We therefore build a controlled experiment on BLN600 (Booth et al., 2024), a corpus of 600 newspaper excerpts with human gold transcriptions. In each of 150 test sentences we damage one block of words around a fact at five nested levels, from one word to 75% of the sentence, with character noise calibrated on the corpus’s own OCR errors. Both

methods receive the same input: the intact context plus the garbled form of each damaged word, so that only the repair differs. We compare two GenAI families: masked infilling with BERT (Devlin et al., 2019), fine-tuned on in-domain gold text and guided by letter similarity in the spirit of “context beats confusion” (Evershed and Fitch, 2014), and instruction-following generation with an open-weight LLM prompted with three solved examples. We score the repaired words with BERTScore (Zhang et al., 2020) for meaning and with a Fact Recovery Rate for exact facts, and we probe the LLM for memorised test text (Sainz et al., 2023). Repair here works on text only: we simulate the OCR output that damage produces, not the physical page, and we assume the damaged region is known; detecting it, and repairing directly from the scanned image with a vision-capable model, are both left to future work.

Our contributions are (i) a fact-centred, graded damage benchmark on BLN600 with nested spans, so the same place gets worse from level to level; (ii) a controlled comparison of masked infilling and few-shot generation under identical input and scoring; and (iii) a fact-level metric next to BERTScore that shows where semantic similarity and factual accuracy part ways. [TODO: Key outcomes, from the TEST split only: (1) the level at which Fact Recovery Rate falls to the no-repair floor; (2) which method wins at low vs high damage, and whether a crossing is a format-failure effect; (3) the size of the BERTScore vs Fact Recovery gap.]

2 Related Work

SQ1: grading damage for repair. No prior study grades damage severity for the repair step itself; graded severity has only been used for *reading* already-OCR’d text. Post-OCR research splits the task into error *detection* and error *correction* (Nguyen et al., 2021), scored separately by the ICDAR 2019 competition (Rigaud et al., 2019); we study correction only and assume the damaged span is known, the assumption Method 3 builds on. Results on how well GenAI corrects disagree: a neural sequence model corrects OCR errors directly (Lyu et al., 2021); a Llama 2 model fine-tuned on BLN600 cut CER further than a fine-tuned BART (Thomas et al., 2024); 14 prompted LLMs were weak across several historical benchmarks (Boros et al., 2024); a seq2seq model beat generative alternatives on early modern Dutch (Debaene et al.,

2025); and prompted LLMs lowered CER for English text but not for Finnish (Kanerva et al., 2025). These results diverge along three lines – fine-tuned versus prompted, the language involved, and how damaged the source corpus already was – but every study reports CER or WER averaged over its corpus’s natural error distribution (Nguyen et al., 2021), which cannot show at what point repair stops working. Graded severity levels do exist, but only for reading OCR text: synthetic degradations at several severities have measured named-entity recognition (Hamdi et al., 2020), and real OCR output has been binned into quality bands for similar downstream tests (Strien et al., 2020). Applying that grading logic to the repair step itself is what SQ1 does.

SQ2: restoring text at known gaps. The closest prior task is restoring damaged ancient text at known positions, and it differs from ours in exactly the ways SQ2 probes. Pythia hides 1–10 characters at known positions in Greek inscriptions and predicts them (Assael et al., 2019); Ithaca extends this and pairs predictions with historians’ own judgement (Assael et al., 2022); a masked-language-model approach fills gaps in Akkadian cuneiform text, with accuracy falling as the masked span grows (Lazar et al., 2021). Three differences motivate our design. First, these studies hide characters completely, whereas OCR damage leaves letters garbled yet partly readable, which our slot input exploits (Sec. 3.3). Second, their gaps are short, at most ten characters or a few tokens, while ours reach 75% of a sentence. Third, none of them compares a masked encoder against a prompted LLM, or scores facts separately from overall similarity – the comparison SQ2 makes. On the model side, BERT fills masked positions in parallel (Devlin et al., 2019), which makes multi-token blanks harder to fill well (Shen et al., 2020) and motivates the short beam search over candidate lengths in Method 3.4; left-to-right generative infilling (Donahue et al., 2020) is the natural counterpart for a prompted LLM; and combining context with letter-level evidence is a classic OCR correction idea (Evershed and Fitch, 2014), of which our reranking step is a direct, modern instance. This also settles whether a masked encoder counts as GenAI at all: BERT can be sampled to produce text (Wang and Cho, 2019), and filling a blank from context is generation conditioned on the rest of the sentence

(Donahue et al., 2020).¹

SQ3: meaning versus facts. A similarity score can pass a repair that gets the facts wrong, and SQ3 tests whether that happens here. BERTScore (Zhang et al., 2020) is the standard similarity metric for judging generated text against a reference. In summarisation it correlates only weakly with human judgements of faithfulness (Spearman 0.19) and factuality (Spearman 0.12) (Maynez et al., 2020), and it can give a high score to a candidate that shares most content words with the reference while stating a different fact (Hanna and Bojar, 2021) – exactly the failure mode of a repair with the right context and one wrong name. This evidence comes from summarisation and machine translation, not from text repair, so SQ3 checks it directly by comparing BERTScore against the Fact Recovery Rate on the same repaired sentences. A second concern is memorisation: BLN600 has been publicly available since 2024, so an instruction-tuned LLM may have seen parts of it during pretraining (Sainz et al., 2023). We probe for this with guided completion: give the model the first part of a test instance and compare its continuation with the true remainder (Golchin and Surdeanu, 2024).

No single element of our design is new by itself; the combination is. Table 1 places our study against the three closest lines of work on five points: whether the damage location is known, whether the damaged letters stay visible, whether severity is graded, whether facts are scored separately from overall similarity, and whether an encoder and an LLM are compared under the same input. The gaps this leaves are what Method 3 builds on: graded nested damage (Sec. 3.2), a shared slot input for both repair methods (Sec. 3.3), two repair families evaluated side by side (Sec. 3.4), and a fact-level score reported next to BERTScore (Sec. 4.2).

3 Method

3.1 Task and Data

Given a sentence whose damaged span is known and whose garbled letters are visible, the task is to predict one word per damaged position; both

¹We cite Wang and Cho (2019) only for this narrow claim; the paper’s own Markov-random-field derivation was later retracted by one of its authors, see <https://kyunghyuncho.me/bert-has-a-mouth-and-must-speak-but-it-is-not-an-mrf/>. A reader unconvinced by the generative framing can instead read BERT as the fine-tuned, in-domain baseline the few-shot LLM has to beat, which is how our experiment treats it either way.

Work	Known loc.	Letters visible	Graded severity	Fact score	Enc. vs LLM
Post-OCR LLMs ^a	×	✓	×	×	×
Pythia ^b	✓	×	×	×	×
Akkadian MLM ^c	✓	×	✓	×	×
Ours	✓	✓	✓	✓	✓

Table 1: Position of this study. ^aThomas et al. (2024); Boros et al. (2024); Kanerva et al. (2025) (real OCR, CER/WER over the corpus). ^bAssael et al. (2019) (1–10 hidden characters). ^cLazar et al. (2021) (masked tokens, results by span length). Graded: results reported by amount of damage.

repair methods take the same input and are scored by the same code (Sec. 3.3). We build this task on BLN600 (Booth et al., 2024), 600 excerpts of London crime reporting from 1830–1890s British newspapers, each with its source image, OCR output and a gold transcription. We split excerpts into sentences with pysbd and pool sentences of 20–60 words with at least two *fact tokens*: a number, any word containing a digit or pound sign, or a capitalised word that is not the first word of the sentence (a short stop list excludes titles such as Mr, Mrs, The). This is a surface rule, not a named-entity tagger: it catches "Old" in "Old Bailey" and misses a lower-cased fact, and it flags dates only through capitalisation or digits, not through date parsing. The pool holds 3,123 sentences from 592 excerpts. From it we draw, with a fixed seed, 150 test and 20 dev sentences from 157 excerpts (at most three per excerpt, no excerpt in both splits, balanced across three sentence-length bands); dev is used only for tuning and is never scored as a result.

3.2 Damage Simulation

For each sentence we draw one anchor among its fact tokens – always damaged, never dropped – and damage one contiguous span of k words that centers on it, with k set so the span covers 1 word, then 10%, 25%, 50% or 75% of the sentence (half-up rounding). Every larger span centers on the same anchor and fully contains every smaller one (Figure 1), so a higher damage level adds more damaged words around a fixed center; it does not make any single word more damaged; the per-word alteration intensity below is drawn the same way regardless of the span size. Each damaged word may be dropped outright (except the anchor). A surviving word has 20–30% of its letters and digits altered, punctuation untouched, split between deletion and substitution as in Table 2 (the substitution

Damage	Type	Value
Word drop	probability	0.10
Letter deletion	probability	0.15 ^a
Letter substitution	probability	0.85
confusion table	share of subst.	85.7%
capitalization	share of subst.	3.6%
look-alike map	share of subst.	4.9%
random character	share of subst.	5.8%

Table 2: Damage operations. Word drop and letter deletion are sampled per word and per altered character (^adeletion capped at 2 per word). Letter substitution is split among four sources by a fixed lookup order (confusion table, then capitalization, then look-alike map, then random), tried until one applies – the “share of subst.” rows are realized frequencies of that lookup, not independent sampling probabilities. The confusion table is calibrated on 29 held-out BLN600 excerpts.

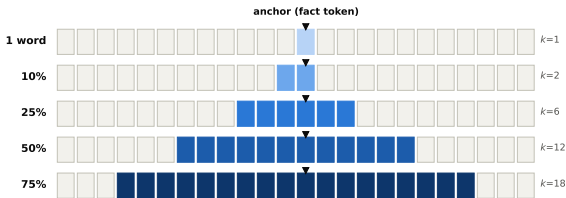


Figure 1: The damaged span at each level, for one example 24-word sentence. Every span centers on the same anchor word (the seeded fact token) and fully contains every smaller span, so a higher level does not change how heavily any single word is damaged – only how many words around the anchor are included.

source is a fixed lookup order – confusion table, then capitalization, then a look-alike map, then a random character – not a random draw between them). The 20–30% band was chosen to sit around the median per-word character error rate we measured on 29 held-out BLN600 excerpts (0.25, IQR 0.17–0.40). All calibration comes from real OCR errors on aligned words, so it models the kind of noise OCR output already contains.

Table 3 shows the result on the test split: sentence-level CER grows roughly in step with real BLN600 OCR error rates, but the share of damaged words that are themselves facts falls from 100% at the 1-word level to 17% at 75%. About one in ten fact slots is dropped outright, leaving no visible letters for that fact at all.

3.3 Shared Slot Input

Both methods see the same *slot view*: the sentence’s undamaged context is left untouched, and each damaged gold word is replaced by a slot showing its garbled form (a dropped word becomes an

empty slot, written $\langle ? \rangle$). This view is built from the gold tokens and the per-word edit operations, never from the raw corrupted text, so it cannot leak an easier version of the task. A predicted word is spliced back inside the original word’s edge punctuation, so context words are never rewritten and every difference in score comes from what a method puts in the slots. For example, the damaged sentence "A POWERFULLY-BUILT man, $\langle \text{same} \rangle$ $\langle \text{Bbonias} \rangle$ $\langle \text{Huicl} \rangle$ $\langle \text{e} \rangle$ commission agent, ..." has the gold continuation "named Thomas Hurd, a"; the garbled forms are visible to both repair methods, but the gold words are not. Showing garbled rather than blanked letters keeps this an OCR-repair task rather than blind text infilling: an OCR engine already flags such regions as low-confidence, and a reader of a damaged page still sees partial letters.

3.4 Repair Methods

(a) **BERT + reranking.** We fine-tune bert-base-cased (Devlin et al., 2019) (110M parameters; cased because fact tokens are partly defined by capitalisation) on clean gold text from the 443 excerpts held outside the sample (2,939 sentences; a further 10% of excerpts held out for validation, with zero excerpt or sentence overlap with test or dev, which we verified directly). Training mirrors inference: for each target word, span words before it are left as gold text, the target is masked with its true number of subword pieces, and later span words each become one [MASK]. Fine-tuning uses a learning rate of $5e-5$, batch size 16, up to 10 epochs with early stopping (patience 2) after a 10%-of-one-epoch warmup and linear decay; the best checkpoint (epoch 2) lowered the per-piece validation loss from 5.61 (pretrained) to 4.14. At inference, slots are filled left to right; for the current slot the model tries 1, 2 and 3 masks, keeps a beam of 5 pieces per position, and retains the top 10 candidates per mask count. A slot with a visible internal hyphen or apostrophe is split into sub-slots filled the same way; only 0.9% of slots (3.5% of fact slots) still need more than three pieces. Each candidate is scored as its mean log-probability per piece plus $\lambda(1 - d)$, where d is the normalised Levenshtein distance to the garbled form, so a fluent completion is pulled back toward what the damaged letters actually show – a direct, modern instance of "context beats confusion" (Evershed and Fitch, 2014). We chose $\lambda = 8$ from a grid over $\{1, 2, 4, 8, 16\}$ on dev, where it gave the best exact-word rate at a point

where Fact Recovery Rate had already plateaued. Dropped slots, which have no garbled letters to match against, are scored on the language-model term alone.

(b) LLM few-shot. The LLM condition prompts an open-weight model with a system message, a task instruction, three solved dev examples (held out from all scoring), and the same slot view as BERT, asking for a JSON list of exactly k strings at temperature 0. *This track was not frozen at the time of writing:* development runs used Llama-3.1-8B-Instruct (Grattafiori et al., 2024) served through GWDG’s SAIA API, with Qwen3-30B-A3B-Instruct as an allowed alternative, but the final model, host and prompt version are being decided by a teammate on separate hardware before the test split is scored – which is also why the Introduction and Related Work name only "an open-weight LLM" rather than a specific model. Only open-weight models hosted this way are used, since BLN600’s CC BY-NC-ND licence rules out sending it to a closed third-party API. The system message itself was necessary: without it, an early check found the model answering in Python or tool-call syntax in 15 of 15 cases rather than plain JSON. An answer of the wrong length, not valid JSON, or cut off by the token limit counts as a *format failure* and is scored as if no repair had been made, with no retry and no manual correction. A secondary variant supplies the whole surrounding article as context instead of just the sentence (its other sentences are gold, not damaged, so this is an upper bound rather than a fair comparison with BERT); we compare it only against the LLM’s own few-shot run. Zero-shot prompting was tried on dev and dropped, so this design cannot isolate how much of the LLM’s performance comes from the three examples themselves rather than the instruction alone.

(c) No-repair baseline. The damaged text itself is scored as if it were a third method, giving the floor both repair methods have to beat. Table 4 summarises both methods’ settings; the LLM row’s model field is left as [TODO: final id] until Sec. 3.4(b)’s open point above is resolved.

4 Empirical Investigation

4.1 Experimental Design

The design crosses three factors inside the same 150 test sentences: five damage levels (1 word, then 10, 25, 50, 75% of the sentence, plus a clean

Level	1w	10%	25%	50%	75%
Sentence CER (%)	1.0	3.2	7.9	15.5	23.1
Facts in span (mean)	1.0	1.6	2.2	3.2	4.2
Facts / damaged words (%)	100	48	26	19	17

Table 3: Damage by level (test split). CER is over the whole sentence against gold; real BLN600 OCR is about 7% (Booth et al., 2024).

<i>BERT + reranking</i>	
Base model	bert-base-cased (110M)
Fine-tuning	lr 5e−5, batch 16, early stop (patience 2, max 10 ep.), best epoch 2
Val. loss	5.61 (pretrained) → 4.14
Reranking	1–3 masks/slot, beam 5, top 10, $\lambda = 8$
<i>LLM few-shot</i>	
Model	[TODO: final id] (dev default: Llama-3.1-8B-Instruct)
Prompt	system msg + 3 dev examples, JSON output
Decoding	temperature 0, max 1024 tokens

Table 4: Implementation settings. λ chosen on dev from {1, 2, 4, 8, 16}.

level 0), three length bands (20–29, 30–44, 45–60 words), and three repair settings (BERT + reranking, LLM few-shot, no repair), with LLM + full article as a secondary check. All five damage levels reuse the same sentence and the same anchor word, so every comparison across levels is paired: for one sentence we can ask directly whether repair got better, worse, or stayed the same as damage grew, rather than comparing averages over different sentences. The 20 dev sentences (disjoint from test, held out from all reported numbers) were used only to pick settings before the test split was scored: the reranking weight λ was chosen on dev alone (Section 3.4), and each method’s setting was frozen by a code gate before its test rows were touched, so nothing reported here was chosen by looking at test results. One setting was still open at the time of writing, as already flagged in Section 3.4: the LLM’s exact model, host and prompt version.

4.2 Metrics

We score the repaired span, not the whole sentence, since only the span was touched. BERTScore F1 (Zhang et al., 2020) compares the repaired span to the gold span using roberta-large (layer 17), rescaled against a baseline so a score can go below 0; this is our primary meaning-level metric. Fact Recovery Rate is our primary fact-level metric: the share of gold fact tokens in the span that the repair restores exactly in their slot, punctuation ignored.

Factor	Levels
Severity	0, 1 word, 10, 25, 50, 75% of words
Length band	20–29, 30–44, 45–60 words
Methods	BERT + rerank; LLM few-shot; no repair (secondary: LLM + article)
Data	150 test sentences, 750 rows per method; dev 20 (tuning only)
Metrics	BERTScore, Fact Recovery Rate (primary); exact match, repair gain, format failures, contamination (secondary)

Table 5: Experimental design.

Level	BERTScore (span)			Fact Recovery Rate		
	BERT	LLM	None	BERT	LLM	None
1w	—	—	—	—	—	—
10%	—	—	—	—	—	—
25%	—	—	—	—	—	—
50%	—	—	—	—	—	—
75%	—	—	—	—	—	—

Table 6: Test results, 150 sentences per level. None = no repair. [TODO: LLM format failures per level; paired counts.]

As secondary metrics we report exact-word match, character error rate (CER, a Levenshtein distance normalised by gold span length) and repair gain (CER before minus CER after), the LLM’s format-failure rate, and a contamination probe (Sainz et al., 2023) that checks whether the LLM simply recites a memorised continuation of a clean test sentence rather than genuinely repairing damaged ones.² A wrong-length LLM answer is a format failure and is scored as no repair, never retried or hand-fixed. BERTScore and Fact Recovery Rate are both reported because they can disagree on purpose (SQ3): a repair that substitutes a wrong but plausible name can still score well on meaning while scoring zero on Fact Recovery Rate, which is exactly the gap we want to measure.

4.3 Results

[TODO: Fill after the LLM test run and notebook 04 on test.]

5 Interpretation and Conclusion

[TODO: Draft after results.]

²Repair gain is reported as a secondary number because it is dominated by short spans: one wrong character in a one-word span already gives CER > 1, so it does not aggregate cleanly across levels; we do not pool it into a single figure.

Limitations

[TODO: Draft limitations as one dense paragraph or a tight list.]

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Appendix Outline (for the defence, not part of this report)

[TODO: One short paragraph listing these items.]